Project Briefs

Software Design | 2026

School of Computer Science and Applied Mathematics

Academic Integrity and Assessment Design

Sprint Review Viva

- At each sprint assessment, the marker will ask individual team members to explain specific parts of the codebase. This includes, for example, walking through a function, justifying a design decision, or modifying a value live during the demo. The ability to answer these questions fluently contributes to the Sprint Review Viva mark.
- Members who cannot explain the code they are presenting will receive zero for the Viva component regardless of the team's overall submission quality.

Individual Sprint Retrospectives

- Each team member must individually submit a short, written retrospective (200-400 words) at the end of every sprint. It must describe what you personally built, what you found difficult, what you would do differently, and how your contribution fits into the overall system. These are submitted individually, not as a group.
- Retrospectives will be cross-referenced with git commit history. Claiming credit for work not reflected in your commits will be treated as academic dishonesty.

Commit History and CI/CD Scrutiny

- Markers will review the full git commit history of your repository. Commits are expected to be incremental, spread across the sprint, and attributable to individual team members. A large spike of commits immediately before a deadline is a red flag.
- Automated CI/CD pipelines must be active throughout the sprint. Build and test history should reflect ongoing development, not a last-minute setup.

Mid-Sprint Requirement Change (Sprint 3)

- At the Sprint 3 planning meeting, a new requirement will be introduced that was not present in the original brief. This is intentional and reflects real-world Agile development. Your team will be expected to incorporate it into your Sprint 3 backlog and demonstrate progress on it by the Sprint 3 assessment.
- This requirement is deliberately not disclosed in advance. It will be scoped to be achievable within the sprint, but will require your team to adapt existing code.

SA Data Integration (highlighted rows)

- Each project includes a mandatory SA Data Integration requirement, highlighted in amber in the requirements table. These rows require your application to source and consume a real, publicly available South African dataset or API relevant to your project domain.
- Finding a suitable data source is part of the requirement. Your team must research available South African public datasets, evaluate their suitability, and document your choice and justification in your project backlog. You will need to understand the data structure, handle real-world data quality issues, and make implementation decisions — this is deliberately not a task that can be fully delegated.

1. Stokvel Management Platform

Introduction

Stokvels are a cornerstone of South African financial culture, with millions of people participating in these rotating savings and credit associations. Managing contributions, tracking payouts, scheduling meetings and maintaining transparency can be challenging when done manually via spreadsheets or messaging apps. This project aims to deliver a web-based stokvel management platform that enables members to track contributions, monitor payout schedules, communicate, and gain financial insights into their savings group.

Objectives

Your team is required to use Agile methodology, incorporate CI/CD principles, and take a test-driven approach to design, develop, and operationalise a publicly available web-based application that meets the following requirements:

Overview of Features

- Member onboarding and group management
- Contribution tracking and payout scheduling
- In-app announcements and meeting management
- Payment integration
- Live interest rate display and savings projections
- Analytics and financial reporting

Detailed Requirements

Requirement	Extra info / Limitations
User Verification	A 3rd party identity provider should be used. 3 roles: Member, Treasurer, Admin.
Group Management	Admins should be able to create and configure stokvel groups, define contribution amounts, payout order, and meeting frequency. Members can be invited and onboarded.
Contribution Tracking	Members should be able to log in and view their contributions. Treasurers should be able to confirm payments, flag missed contributions and manage the payout schedule.
Meeting Management	Treasurers and Admins should be able to schedule meetings, post agendas, and record minutes. Members should receive notifications for upcoming meetings.
Payments	Integrated with a 3rd party payment gateway. Members should be able to make contributions online. Treasurers should be able to initiate payout disbursements.
SA Data Integration	The application must display the current South African prime lending rate and repo rate, calculated using live or regularly updated public data. Your team must research and identify a suitable publicly available South African data source for this information. You must document your chosen source and justify why it is reliable in your project backlog. Members' projected savings growth must be derived from this data.
Analytics	At least 3 dashboard reports: • Contribution compliance per member over time

Requirement	Extra info / Limitations
	• Payout history and upcoming payout projections • Custom view Reports should be exportable as CSV or PDF.

Bonus: *ML-powered financial health scoring for individual members based on their contribution history and engagement patterns.*

2. SA Township Business Connect

Introduction

South Africa's township economies are vibrant and entrepreneurial, yet many small and micro businesses lack a digital presence that would allow them to reach customers beyond their immediate neighbourhood. Potential customers also struggle to discover township-based goods and services. This project aims to build a web-based directory and e-commerce platform specifically tailored for township and informal businesses, empowering them to list their products or services, accept orders, and grow their customer base.

Objectives

Your team is required to use Agile methodology, incorporate CI/CD principles, and take a test-driven approach to design, develop, and operationalise a publicly available web-based application that meets the following requirements:

Overview of Features

- Business profile and product/service listing
- Customer browsing, search, and ordering
- Municipality-aware location filtering
- Order and delivery management
- Payment integration
- Analytics and reporting

Detailed Requirements

Requirement	Extra info / Limitations
User Verification	A 3rd party identity provider should be used. 3 roles: Business Owner (Seller), Customer (Buyer), Admin.
Business Profiles	Business Owners should be able to create a profile with a name, location, description, operating hours, and photos. Admins can approve or remove listings.
Product & Service Listings	Business Owners should be able to list products or services with prices, descriptions, photos, and availability. Stock levels should be manageable.
Customer Experience	Buyers should be able to browse and search by category, location, or keyword and place orders. Buyers should be able to leave ratings and reviews after a transaction.
Payments	Integrated with a 3rd party payment gateway. Buyers should be able to pay online at the time of ordering. Business Owners should be able to track earnings.
SA Data Integration	Business location search and filtering must use real South African geographic boundary data to tag each business listing to a ward and municipality. Your team must research and identify a suitable publicly available source for SA ward or municipal boundary data (e.g. GeoJSON or shapefile format). You must document your chosen source and justify its suitability in your project backlog.
Analytics	At least 3 dashboard reports: • Sales trends per business over time • Top-performing product/service categories

Requirement	Extra info / Limitations
	• Custom view Reports should be exportable as CSV or PDF.

Bonus: *AI-powered product recommendations for buyers based on browsing history and location.*

3. SA Learnerships and Skills Development Portal

Introduction

South Africa faces a significant youth unemployment challenge, with many young people unaware of or unable to access SETA-accredited learnerships, apprenticeships, and internship opportunities. At the same time, employers and training providers struggle to reach suitable applicants efficiently. This project aims to create a web-based platform that connects work-seekers with learnerships and skills development opportunities, streamlining the application and tracking process for both applicants and providers.

Objectives

Your team is required to use Agile methodology, incorporate CI/CD principles, and take a test-driven approach to design, develop, and operationalise a publicly available web-based application that meets the following requirements:

Overview of Features

- Opportunity listings (learnerships, internships, apprenticeships)
- Applicant profiles and NQF-aligned skill tagging
- Application review and selection workflow
- Notifications and status tracking
- Analytics and reporting

Detailed Requirements

Requirement	Extra info / Limitations
User Verification	A 3rd party identity provider should be used. 3 roles: Applicant, Provider (Employer / Training Organisation), Admin.
Opportunity Listings	Providers should be able to post learnership or internship opportunities, including title, description, stipend, location, duration, requirements, and closing date. Admins can approve or remove listings.
Applicant Profiles	Applicants should be able to create a profile with personal details, education, skills, and upload a CV. Applicants should be able to apply for multiple opportunities and track the status of each application.
Application Workflow	Providers should be able to view, shortlist, and reject applications. Applicants should receive notifications at each stage (received, shortlisted, rejected, offered).
Notifications	Users should receive in-app and email notifications for application status updates, new matching opportunities, and upcoming closing dates.
SA Data Integration	All qualifications and skill tags on applicant profiles and opportunity listings must be aligned to the South African National Qualifications Framework (NQF). Your team must research and identify a publicly available source for valid NQF levels and registered qualification types (e.g. from SAQA or a related government body). The application must use this data to populate qualification dropdowns rather than hardcoded values. Document your chosen source and justify it in your project backlog.
Analytics	At least 3 dashboard reports: • Application volume per opportunity • Placement success rates by sector

Requirement	Extra info / Limitations
	• Custom view Reports should be exportable as CSV or PDF.

Bonus: *AI-based opportunity matching that recommends relevant learnerships to applicants based on their profile and skills.*

4. Community Clinic Appointment and Queue Management System

Introduction

Public health clinics across South Africa are frequently overcrowded, with patients often waiting for several hours with no visibility into their position in the queue. This leads to frustration and, in some cases, patients leaving before receiving care. This project aims to deliver a web-based appointment and queue management system for community clinics, allowing patients to book appointments, join virtual queues remotely, and receive real-time updates, while enabling clinic staff to manage patient flow efficiently.

Objectives

Your team is required to use Agile methodology, incorporate CI/CD principles, and take a test-driven approach to design, develop, and operationalise a publicly available web-based application that meets the following requirements:

Overview of Features

- Appointment booking and queue management
- Real-time patient status tracking
- Clinic directory sourced from the national health facility register
- Clinic and staff management
- Notifications and reminders
- Analytics and reporting

Detailed Requirements

Requirement	Extra info / Limitations
User Verification	A 3rd party identity provider should be used. 3 roles: Patient, Clinic Staff, Admin.
Appointment Booking	Patients should be able to view available appointment slots at their nearest clinic and book, reschedule, or cancel appointments. Walk-in patients should be able to join a virtual queue on arrival.
Queue Management	Clinic Staff should be able to manage the daily patient queue, update patient statuses (Waiting, In Consultation, Complete), and add or reschedule patients as needed.
Patient Status Tracking	Patients should be able to view their estimated wait time and position in the queue in real time. Notifications should be sent when their turn is approaching.
SA Data Integration	The clinic directory must be seeded from a real, publicly available South African health facility dataset. Your team must research and identify a suitable source (e.g. a government open data portal or the National Department of Health). The dataset must include real clinic names, locations, facility types, regions, and provinces. Clinic search must support filtering by facility type, services offered, regions, province, and district using this data. Document your chosen source and justify its reliability in your project backlog.
Clinic Management	Admins should be able to manage operating hours and services offered for each registered facility and assign staff. Staff should be able to set their availability.

Requirement	Extra info / Limitations
Analytics	At least 3 dashboard reports: • Average patient wait times by clinic and time of day • Appointment no-show rates • Custom view Reports should be exportable as CSV or PDF.

Bonus: *ML model for predicting patient wait times based on historical queue data, time of day, and day of the week.*

5. Municipal Service Delivery Reporting Portal

Introduction

Poor municipal service delivery is a persistent challenge in South Africa, with residents struggling to report issues such as potholes, burst pipes, illegal dumping, and power outages to their local municipalities. Reports made via phone or social media are often lost or unacknowledged. This project aims to build a web-based service delivery reporting portal where residents can submit, track, and escalate service requests, while municipal staff manage and resolve issues in an accountable and transparent manner.

Objectives

Your team is required to use Agile methodology, incorporate CI/CD principles, and take a test-driven approach to design, develop, and operationalise a publicly available web-based application that meets the following requirements:

Overview of Features

- Service request submission with ward-level geolocation
- Request assignment and resolution workflow
- Resident notifications and feedback
- Public service delivery dashboard
- Analytics and reporting

Detailed Requirements

Requirement	Extra info / Limitations
User Verification	A 3rd party identity provider should be used. 3 roles: Resident, Municipal Worker, Admin.
Service Request Submission	Residents should be able to submit a service request with a category (e.g. potholes, water, electricity, waste), description, photo, and geolocation. Residents should be able to view and track all their submitted requests.
SA Data Integration	Service requests must be automatically tagged to the correct ward and municipality based on the location submitted by the resident. Your team must research and identify a publicly available South African ward or municipal boundary dataset (e.g. in GeoJSON or shapefile format). The submission map must render real ward boundaries and municipality names from this data. Document your chosen source and justify its suitability in your project backlog.
Request Management	Municipal Workers should be able to view, claim, update, and resolve assigned requests. Admins should be able to assign requests to workers and set priority levels.
Notifications & Feedback	Residents should receive notifications when their request status changes (e.g. Acknowledged, In Progress, Resolved). Residents should be able to submit satisfaction feedback once a request is resolved.
Public Dashboard	A publicly visible, read-only dashboard showing the status and location of open and recently resolved service requests in the community, rendered on a ward boundary map. No login required.
Analytics	At least 3 dashboard reports: • Request volume and resolution times by category

Requirement	Extra info / Limitations
	• Worker performance (requests resolved per period) • Custom view Reports should be exportable as CSV or PDF.

Bonus: *AI-driven automatic categorisation and priority scoring of incoming service requests based on issue description and historical patterns.*

6. Campus Marketplace

Introduction

University students may want to buy, sell, or trade new or used items such as textbooks, electronics, furniture, and clothing. Existing general-purpose marketplaces lack university-specific convenience and trust mechanisms, making it difficult to make purchases through them. This project aims to deliver a web-based peer-to-peer trading platform exclusively for university students, enabling them to list, discover and safely exchange or trade new and used items within their campus communities. Each campus is set to have a secure trade management facility that allows students to safely drop off and pick up trade items. All payments are to be made online, and any trade shortfall can be supplemented by cash payment through the same online payment platform.

Objectives

Your team is required to use Agile methodology, incorporate CI/CD principles, and take a test-driven approach to design, develop, and operationalise a publicly available web-based application that meets the following requirements:

Overview of Features

- Item listing and browsing system
- Secure trade facility management (drop-off and collection)
- Built-in messaging system
- Online payments with cash shortfall supplement
- Rating and trust system
- Analytics

Detailed Requirements

Requirement	Extra info / Limitations
User Verification	A 3rd party identity provider should be used. 3 roles: Student, Trade Facility Staff, Admin.
Listing Management	Students should be able to create, edit, and remove item listings with photos, descriptions, categories, condition ratings, and asking prices. Listings should indicate whether the item is for sale, trade, or either.
Buyer Experience	Buyers should be able to search, filter by category, condition, and price range, and initiate a purchase or trade offer. Buyers should be able to view a seller's rating and transaction history before committing.
Trade Facility Management	Trade Facility Staff should be able to manage drop-off and collection bookings for items involved in a transaction. Students should be able to book a drop-off time slot online. Staff should confirm item receipt and release, and update transaction status accordingly. Admins should be able to configure facility operating hours and slot capacity.
In-App Messaging	Buyers and sellers should be able to communicate through a built-in chat to negotiate and arrange transaction details before committing to a trade facility booking.
Payments	Integrated with a 3rd party payment gateway. Students should be able to pay the full amount or a partial amount online, with any cash shortfall recorded and tracked through the platform. Trade facility staff should be able to confirm that any outstanding cash component has been settled before releasing items.

Requirement	Extra info / Limitations
Rating & Trust System	After a transaction is marked complete, both parties should be able to leave ratings and reviews. Admins should be able to flag or remove abusive content.
SA Data Integration	Item price suggestions should be informed by publicly available South African consumer price data relevant to the item category (e.g. electronics, textbooks, furniture). Your team must research and identify a suitable publicly available SA data source for this purpose. Document your chosen source and justify its suitability in your project backlog.
Analytics	At least 3 dashboard reports: • Most popular item categories and completed transactions over time • Trade facility utilisation (slot bookings vs capacity) • Flagged or moderated content summary Reports should be exportable as CSV or PDF.

Bonus: *ML model integration for item price suggestions based on category, condition, and recent comparable sales on the platform.*

7. Campus Food Ordering Platform

Introduction

University food courts (e.g. shops within the Matrix) often have long queues during peak hours, which can cause students to miss classes or skip meals. At the same time, vendors lack a centralised system for managing orders and monitoring demand. This project aims to create a web-based food ordering platform for a university campus, enabling students to browse menus, place orders online and collect their food with minimal wait time.

Objectives

Your team is required to use Agile methodology, incorporate CI/CD principles, and take a test-driven approach to design, develop, and operationalise a publicly available web-based application that meets the following requirements:

Overview of Features

- Multi-vendor menu browsing and ordering
- Order queue and status tracking
- Payment processing
- Vendor management
- Analytics

Detailed Requirements

Requirement	Extra info / Limitations
User Verification	A 3rd party identity provider should be used. 3 roles: Student (Customer), Vendor, Admin.
Menu Management	Vendors should be able to create and update their menus, including item names, descriptions, prices, photos, and availability. Items should be markable as sold out. Admins should be able to approve or suspend vendor accounts.
Order Management	Students should be able to place orders from one or more vendors. Vendors should receive and manage incoming orders, update preparation status, and mark orders as ready for collection.
Order Tracking	Students should receive real-time status updates (e.g. Order Received, Preparing, Ready for Pickup). Notifications should be sent when the order is ready for collection.
Payments	Integrated with a 3rd party payment gateway. Students should be able to pay online at the time of ordering.
SA Data Integration	Menu items must include allergen and dietary information (e.g. halal, vegan, nut-free). Your team must research and identify a publicly available South African or internationally recognised dietary/allergen data source to standardise this information across vendors. Document your chosen source and justify it in your project backlog.
Analytics	At least 3 dashboard reports: • Sales per vendor over time • Peak ordering hours • Custom view Reports should be exportable as CSV or PDF.

Bonus: *ML model integration for meal recommendations based on student order history and dietary preferences or restrictions.*